

Ex.No.8 Use Steg-Expose to detect hidden data in images

Date:

Description

StegExpose is a tool used to detect hidden data in images using steganography analysis. It works by evaluating the statistical properties of an image to estimate whether any hidden data is embedded. Below is a step-by-step guide on how to use StegExpose to detect hidden data in images:

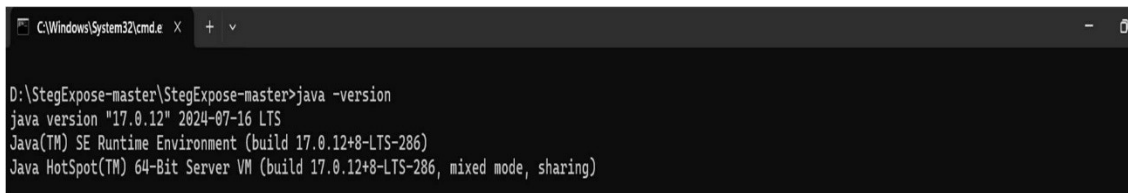
Prerequisites:

1. Java Runtime Environment (JRE): StegExpose is a Java-based tool, so ensure you have JRE installed.
2. StegExpose Tool: Download StegExpose from the [official GitHub repository](#).

Step-by-Step Process:

1. Download and Set Up StegExpose:

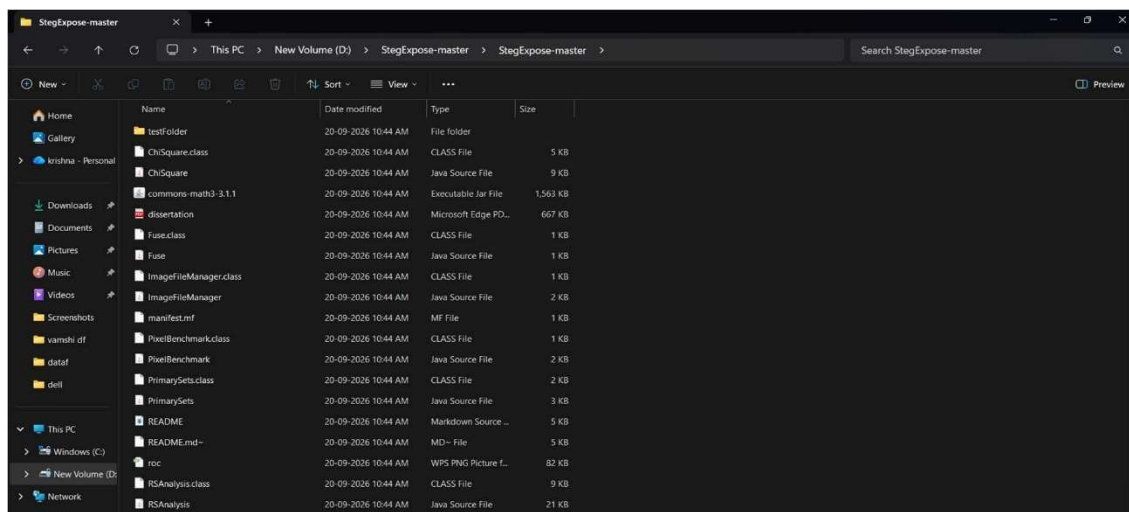
- Download the tool: Go to [StegExpose GitHub](#) and download the .jar file.
- Install Java: Ensure that Java is installed on your machine. You can download it from the official site.



```
C:\Windows\System32\cmd.exe X + v

D:\StegExpose-master\StegExpose-master>java -version
java version "17.0.12" 2024-07-16 LTS
Java(TM) SE Runtime Environment (build 17.0.12+8-LTS-286)
Java HotSpot(TM) 64-Bit Server VM (build 17.0.12+8-LTS-286, mixed mode, sharing)
```

- Prepare environment: Place the .jar file in a folder where you plan to work on the image files.
- 2. Select Images for Analysis:
- Collect the images you suspect might have hidden data. StegExpose supports formats such as .png, .jpg, .bmp, etc.



3. Open Command Line or Terminal:

- Navigate to the folder where you have saved the StegExpose.jar file.
- Open your Command Prompt (Windows) or Terminal (Linux/macOS).

4. Run StegExpose on an Image:

- Use the following command to check a single image for hidden data:

bash

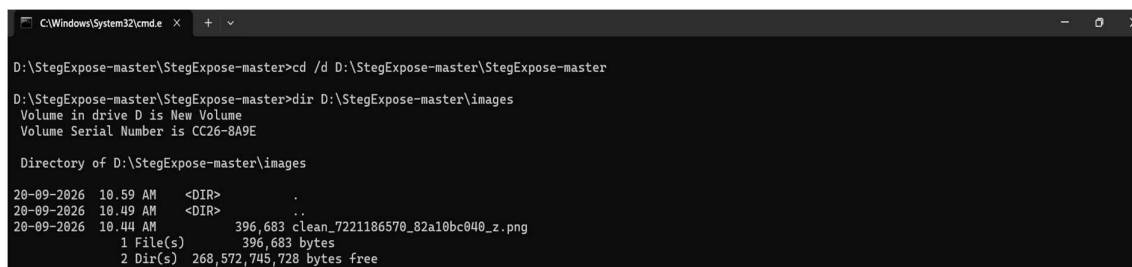
```
java -jar StegExpose.jar <image_file_path>
```

Example:

bash

```
java -jar StegExpose.jar test_image.png
```

- Replace <image_file_path> with the actual path of the image you want to check.



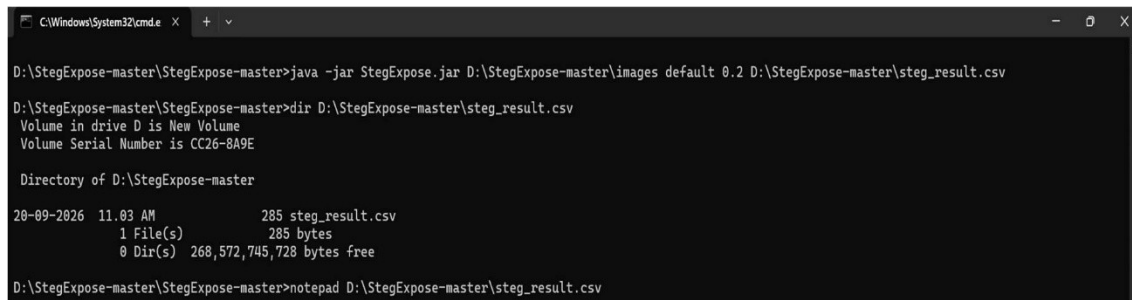
```
C:\Windows\System32\cmd.exe
D:\StegExpose-master\StegExpose-master>cd /d D:\StegExpose-master\StegExpose-master
D:\StegExpose-master\StegExpose-master>dir D:\StegExpose-master\images
Volume in drive D is New Volume
Volume Serial Number is CC26-8A9E

Directory of D:\StegExpose-master\images

20-09-2026  10:59 AM  <DIR>          .
20-09-2026  10:49 AM  <DIR>          ..
20-09-2026  10:44 AM                396,683 c\lean_7221186570_82a10bc040_z.png
               1 File(s)                396,683 bytes
               2 Dir(s)      268,572,745,728 bytes free
```

5. Analyze the Output:

- After running the tool, it will analyze the image and provide a score.
- Output Explanation:
 - o StegExpose calculates a "suspect" score (between 0 and 1). The higher the score, the more likely that hidden data is present.
 - o Threshold values (suggested by the tool):
 - ☐ Less than 0.2: Image is clean (no hidden data).
 - ☐ 0.2 - 0.3: Possibly some hidden data.
 - ☐ Above 0.3: Likely that steganography is present.



```
C:\Windows\System32\cmd.exe
D:\StegExpose-master\StegExpose-master>java -jar StegExpose.jar D:\StegExpose-master\images default 0.2 D:\StegExpose-master\steg_result.csv
D:\StegExpose-master\StegExpose-master>dir D:\StegExpose-master\steg_result.csv
Volume in drive D is New Volume
Volume Serial Number is CC26-8A9E

Directory of D:\StegExpose-master

20-09-2026  11:03 AM                285 steg_result.csv
               1 File(s)                285 bytes
               0 Dir(s)      268,572,745,728 bytes free

D:\StegExpose-master\StegExpose-master>notepad D:\StegExpose-master\steg_result.csv
```

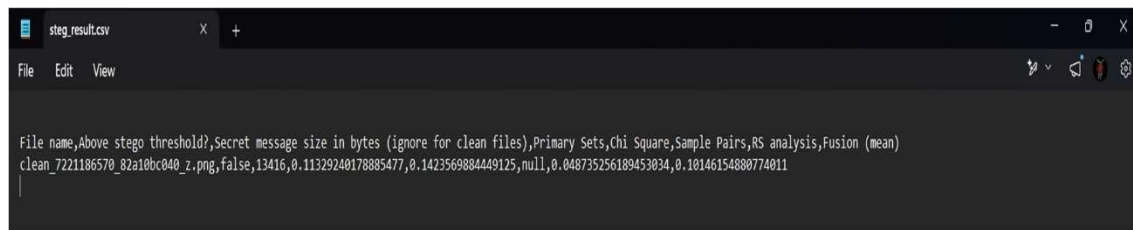
6. Batch Analysis (Multiple Images):

- If you have multiple images to check, you can analyze them in bulk by specifying a folder.
For example:

```
bash
```

```
java -jar StegExpose.jar <folder_path>
```

- Replace <folder_path> with the path to the folder containing the images.



```
File name,Above stego threshold?,Secret message size in bytes (ignore for clean files),Primary Sets,Chi Square,Sample Pairs,RS analysis,Fusion (mean)
clean_7221186570_82a10bc040_z.png,false,13416,0.11329240178885477,0.1423569884449125,null,0.048735256189453034,0.10146154880774011
```

7. Advanced Options (Optional):

- StegExpose provides additional parameters, such as adjusting the sensitivity of detection and output verbosity. Use --help to see the list of options:

```
bash
```

```
java -jar StegExpose.jar --help
```

8. Review the Results:

- For each image analyzed, review the scores and determine whether you need to investigate further.

Example:

```
bash
```

```
java -jar StegExpose.jar
```

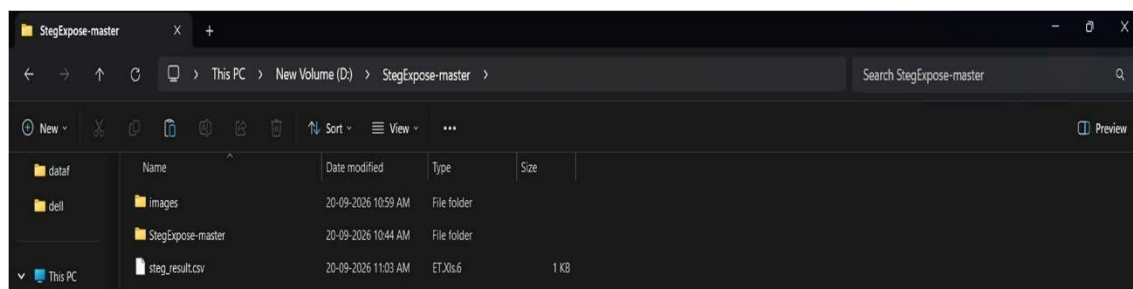
```
suspect_image.png Output: makefile
```

```
Analyzing suspect_image.png...
```

```
Result: 0.4
```

```
Steganography likely present
```

This output indicates that hidden data is likely present in the image suspect_image.png.



Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

StegExpose was successfully used to analyze the image clean_7221186570_82a10bc040_z.png. The image obtained a Fusion (mean) score of 0.10146, which is below the threshold of 0.2. Therefore, the image was classified as clean, with no likely hidden steganographic data detected.